

G Tonewheel

A tonewheel drawbar organ with percussion, key click, scanner vibrato / chorus and valve drive.



Overview

G Tonewheel is a drawbar organ in the classic tonewheel style. 91 tonewheels spin all the time and stay locked together, as on the real thing; nine drawbars from 16' to 1' mix their harmonics, with the classic foldback at the top and bottom of the range.

It has the parts that make the sound: percussion on the 2nd or 3rd harmonic (single-trigger - it only strikes again once all keys are lifted), the key click of the contacts, the scanner vibrato and chorus (V1 - V3, C1 - C3), a valve preamp DRIVE, TONE and the SWELL pedal.

The organ is mono and dry by design. For the full sound, put GFX Rotary after it: G Tonewheel passes all MIDI on, so the mod wheel or sustain pedal can switch the rotary's speed.

Reach for it for

- Jazz, soul and gospel organ parts
- Rock organ with overdrive (into GFX Rotary)
- Reggae bubble and other percussive comping
- Soft flute and ballad registrations
- Swell-pedal dynamics from an expression pedal (CC11)

Playing it

G Tonewheel is an instrument: it has no audio input and makes its sound from MIDI.

1. Insert G Tonewheel on a track (as the first plug-in, or after a MIDI effect).
2. Arm the track for recording and set its input to your MIDI keyboard (or draw notes in a MIDI item).
3. Pull the drawbars, or pick a registration from PRESETS.
4. Play. The MIDI IN light shows notes arriving, and the MIDI section shows how many keys are down.
5. For the classic rotary-speaker sound, add GFX Rotary after G Tonewheel on the same track.

G Tonewheel responds to the expression pedal (CC11 moves SWELL) and all-notes-off (CC120 and CC123). Velocity does not change the level, as on a real organ. All MIDI is passed on down the chain.

Controls

The header

PRESETS	Opens the list of drawbar registrations. Choosing one sets the drawbars, percussion, vibrato / chorus, CLICK and DRIVE; TONE, SWELL, TUNE, DRIFT and OUTPUT are left as they are. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to

close it.

PANIC

Lets go of every held key at once.

DRAWBARS

Nine drawbars, left to right: 16', 5 1/3', 8', 4', 2 2/3', 2', 1 3/5', 1 1/3' and 1'. The two brown ones are the sub-harmonics, the white ones the octaves of the note and the black ones the fifths and the third. Drag a drawbar down to pull it out (0 to 8), or use the mouse wheel over it. Each step is about 3 dB; 0 is off.

PERC

ON

Switches the percussion on: a decaying ping on the 2nd or 3rd harmonic at the start of each phrase. It is single-trigger: it only strikes again once all keys are lifted, so play detached for percussion on every chord. While it is on, the 1' drawbar is silent (the percussion borrows it).

THIRD

Off: percussion on the 2nd harmonic (an octave up). On: on the 3rd harmonic (an octave and a fifth up).

SOFT

Off (NORMAL): loud percussion, and the drawbars drop by about 3 dB to make room for it. On: softer percussion, the drawbars at full level.

SLOW

Off (FAST): a short ping. On: a longer decay.

THIRD, SOFT and SLOW only work (and only light up) while ON is lit.

V/C

V1 / V2 / V3

Scanner vibrato, from shallow (V1) to deep (V3).

C1 / C2 / C3

Scanner chorus: the vibrato mixed with the straight sound, from shallow to deep.

Off

Click the lit key again to switch vibrato and chorus off.

VOICE

CLICK	The key click, 0 - 100% (40% by default): how fast the contacts switch on and off, plus a little contact noise. Turn it down for a smooth, soft attack; up for a percussive spit.
DRIVE	The valve preamp, 0 - 100% (15% by default). Low settings add warmth; high settings break up into growl.
TONE	Brightness, 0 - 100% (70% by default): a low-pass that rolls the top off, from about 2 kHz at 0% to 12 kHz at 100%.
SWELL	The expression pedal, 0 - 100%. It follows a squared curve, like a real swell pedal. MIDI CC11 moves it.
TUNE	Fine tuning, -50 to +50 cents.

Drift

DRIFT	0 - 100% (25% by default). The motor speed wanders slowly, so the whole organ drifts a little in pitch (up to 20 cents at 100%), TONE drifts a little brighter and darker, and the scanner's speed wavers. The tonewheels always stay locked together, so the organ never goes out of tune with itself.
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MIDI

CHANNEL	The MIDI channel G Tonewheel listens to: OMNI (all) or CH 1 to CH 16.
MIDI IN	Lights when a note arrives or keys are held. Below it, the number of keys down.

MASTER

OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right, with a floating peak. Click a meter to clear the peaks.

The output is softly clipped at -0.3 dBFS, so peaks never go over.

Presets

Soul Jazz	888000000 with third-harmonic percussion (normal, fast) and C3.
Memphis Groove	888800000, no percussion, C3 and a little more drive.
Full Organ	All drawbars out, C3, more drive.
Gospel	886800008 - the 1' on top - with C3 and more drive.
Modern Jazz	838000000 with soft, slow third-harmonic percussion, no vibrato and almost no drive.
Jazz Ballad	808000000, gentle C1 chorus, a soft click.
Rock Overdrive	888888000, C3, a sharper click and heavy drive.
Reggae Bubble	008800000 with soft, fast second-harmonic percussion and no vibrato.
Flute	The 8' alone with V1 vibrato, a very soft click and no drive.
Slow Perc	888000000 with slow third-harmonic percussion and C3.
Reset all to defaults	Every control back to its starting value.

Tips

- Drawbar numbers read left to right: 888000000 means the first three drawbars fully out.
- For percussion on every chord, lift all keys between chords; held legato notes will not re-strike it.
- Put GFX Rotary straight after G Tonewheel and use the mod wheel or sustain pedal to switch it between slow and fast.
- Map an expression pedal to CC11 for SWELL: it is the most expressive control on an organ.
- If the organ sounds harsh with lots of upper drawbars, bring TONE down rather than pushing the drawbars back in.

File: gfx-g-tonewheel.jsfx